

Extraction and Access to Information in Natural Language for Non-Developers - Democratizing Information

Progress Report – 2022/2023
2/2 Partial Time

Emanuel Matos



University of Aveiro
Portugal
September - 2023

Contents

1	Introduction	1
2	Developments	2
2.1	Dataset(s)	3
2.1.1	Dataset - WikiNER	3
2.1.2	Initial datasets	3
2.2	Test of BERT-based NER in new datasets	5
2.3	Exploration of state-of-the art Algorithms	6
2.3.1	BERT	6
2.3.2	GPT Family	7
2.4	Initial experiments with Open IE based NER	7
3	Results	8
3.1	Publications and Presentations	8
3.1.1	IberSpeech 2022	8
3.1.2	New paper in preparation	9
3.1.3	Doctoral Consortium	9
3.2	New pipeline using Open IE	11
3.3	Comments	14
4	Future work	15
4.1	Updated Workplan	16
4.1.1	Milestones - Update	16
5	Conclusion	18

Chapter 1

Introduction

Automatic Information Extraction (IE) from natural language sources has many applications, including Business Intelligence [1], Forensics [2], Medicine [3] and question answering systems [4]. Named Entity Recognition (NER) has a very important role in IE. As development of NER for new domains is highly dependent of annotated data, the main objective of our work is to contribute with methods enabling NER for new domains lacking annotated datasets. Seeking to work new concepts and diversity, both in terms of the environment, various datasets, such as the technical issue and experiments, the year of 2021 was very interesting.

Even with the limitation of working part-time and being in Brazil, we were able to immerse ourselves in a world of new algorithms and evaluate diverse possibilities, as well as have technical production that allowed us to publish and present at an international conference. Also, the results obtained during this process of experimentation led us to develop a pipeline that will be the basis for the next year.

This report describes the work developed during the 3rd year of Thesis work ¹.

¹I'm a part-time student, this 3rd year corresponds to the second half of the 2nd year of Thesis.

Chapter 2

Developments

The main developments during the third year of Thesis work were:

- At the beginning of the year, in the 1st quarter, I participated in the Iber-SPEECH 2022 congress, which took place in November in Granada. I prepared and presented a poster on the submitted and published communication.
- In December we started creation of a new dataset to support creation of new models, to be used to continue evaluation of the hypothesis raised in the initial years, that is to create NER systems without manually annotated data.
- In the second quarter we exhaustively tested the modeling generated by BERT. The results obtained were below expectations, but on the other hand we developed a dataset, we organized the database that will be an added value in the work that we will be able to carry out.
- Improvements to the NERs previously developed and their output combination.
- Publication of results.

In this year, there were some acquisition of knowledge and state-of-the-art update.

The next subsections present additional information on the main developments:

2.1 Dataset(s)

This last year we continued to work with our initial datasets and introduced a new dataset (WikiNER), with the aim of creating domain diversity. This is necessary to answer questions about the theory/process we are developing.

2.1.1 Dataset - WikiNER

Created by Nothman[5], the WikiNER dataset contains 7.200 manually labeled Wikipedia articles in nine multilingual languages: English, German, French, Polish, Italian, Spanish, Dutch, Portuguese, and Russian. This year we used this dataset as part of our strategy to diversify assessment environments.

The dataset includes manually annotated entities that we only use for evaluation purposes. The set of annotated entities is :

- LOC, example: Rivers of, Towns
- ORG, example: Organizations, musical groups
- PER, example :Living People, Year of birth
- MISC, example: Television series, discographies
- NON, example: Years, Wikipedia
- DAB, example: Disambiguation

WikiNER was chosen because it has a large volume of free multilingual training notes for named entity recognition (NER), exploring the text and structure of Wikipedia. And it includes Portuguese, which is our language of study.

2.1.2 Initial datasets

Next we briefly describe the other datasets that we maintained from previous work.

Tourism Domain

For our initial proof-of-concept the area of Tourism was selected. A set of sources were selected manually, and the text scrapped using the Scrapy library for retrieving documents and BeautifulSoup library for getting document data.

The option for Tourism domain resulted from our previous work in this domain [6]. Selected by the high potential of automatic information extraction to provide relevant information to several domain stakeholders (e.g., hotel managers).

The main source selected was Wikivoyage [7], with hundreds of texts regarding countries, regions, cities, tourist attractions, etc. Based on the tool's ease of use concept, the manual extraction option, without advanced features, was implemented to adjust expectations regarding the development of the work and evaluate the capture properties of possible entities without advanced techniques.

LeNER Legal texts dataset

LeNER-Br is a Portuguese language dataset for named entity recognition applied to legal documents. LeNER-Br consists entirely of manually annotated legislation and legal cases texts and contains tags for persons, locations, time entities, organizations, legislation and legal cases. To compose the dataset, 60 legal documents from several Brazilian Courts were collected. Courts of superior and state levels were considered, such as Supremo Tribunal Federal, Superior Tribunal de Justiça, Tribunal de Justiça de Minas Gerais and Tribunal de Contas da União. In addition, four legislation documents were collected, such as “Lei Maria da Penha”, giving a total of 70 documents.

Paramopama

Extends the PtBR version of WikiNER corpus, revising incorrect assigned tags in order to improve corpus quality, also extend the corpus size and provide proper evaluation[8]. This dataset has a total of 240,755 words tagged as part of an entity, considering 4 types (PERSON, LOCATION, ORGANIZATION and TIME).

2.2 Test of BERT-based NER in new datasets

Despite recent advances, particularly using deep learning techniques, creating accurate named entity recognizers remains a complex task, highly dependent on the availability of annotated data. To advance the existence of NER systems for new domains, it is crucial to obtain the necessary large volumes of annotated data with little or no manual work. Aligned with this, between January and April, we tested BERT-trained model that we consider to be a good way to search for efficient NERs.

In our published work[9] we evaluated a new dataset with the process described in the article [10] so that we have more information to evaluate the generalization of this modeling. The results obtained indicated that the method has some generalization potential. Then we started evaluating the creation of a single entity for Modeling (i.e., annotating word as belonging or not to an ENTITY). This approach was tested on a dataset of approximately 121,000 sentences extracted from WikiNER. The results obtained did not meet the expectations created. There were no good results in the traditional evaluation metrics such as Precision, Recall and F1.

The results per class that we got after an exhaustive pre-processing are presented in Figure 2.1 The figures represent the confusion matrix between Single Entity BERT model output and WikiNER's manual annotation (marked as ORIGINAL in the tables) for 3 types of entities. It is clear the low performance, with most of ORG, LOC not detected. The corresponding results in terms of Precision, Recall and F1 are presented in Table 2.1.

Table 2.1: Comparative Results of Unique Entity Modelling.

Indicator	ORG	LOC	PER	MISC
Precision	0.31	0.10	0.21	0.89
Recall	0.10	0.00	0.00	0.11
F1	0.15	0.00	0.00	0.21

A major issue that we had to address during these tests was the comparison between the ground truth (manual annotations) and the output of our models. The large number of sentences and the alignment of the different tokenizations required an expected investment in terms of time.

BERT	O	ORG
ORIGINAL		
O	3540611	9415
ORG	36840	4233

(a) NER:ORG

BERT	LOC	O
ORIGINAL		
LOC	31138	198041
O	3532	3358388

(b) NER:LOC

BERT	MISC	O
ORIGINAL		
MISC	51599	12259
O	420094	3107147

(c) NER:MISC

Figure 2.1: Confusion matrices obtained for 3 classes (ORG, LOC, MISC) in experiments with one-class BERT-base classifiers.

The experiments also made clear that the number of classes worked does not reflect the real world well. The used datasets only have a trivial set of classes such as Local, Organization, Person and Miscellaneous, while, for example, our dbpedia-based NER uses a much larger and diverse dataset. This motivated us with the search for a new NER that would allow us to have diversity in the initial automatic annotation.

2.3 Exploration of state-of-the art Algorithms

An active search for state-of-the-art developments was carried out to select among the most relevant in terms of cost, performance and result, with a focus on the 'democratization of knowledge'. Our initial option was BERT, but we are still evaluating the 'spin offs' of this algorithm, like RoBERTa and others.

2.3.1 BERT

The implementation we have been using is based in BERT[11] implementation by Tobias Sterbak using the Transformers package by Huggingface [12], Keras and

TensorFlow.

Last year we started to address the train of ML systems using data annotated automatically. Not just expressing data in the format(s) accepted by existing systems but also doing so in a way non-developers feel comfortable with.

Aiming to mitigate the problems highlighted above, the following main objectives were adopted for the ongoing work on this topic:

1. Develop processes to simplify the creation of NER systems for new domains, starting by the creation of the needed annotated data;
2. Make NER deployment as easy as possible in order to be used by non specialists, contributing to breaking existing usage barriers thus fostering wider adoption of such systems.

2.3.2 GPT Family

During the months of October to early January 2023, we evaluated the possibility of using GPT3, GPT2, GPT-J and GPT-Neo.

We looked for and evaluated free alternatives and with reasonable demands in terms of requirements, as one of the basic issues of this work is to be democratic, and with costs that means restriction.

GPT-Neo was the one that came closest to the possibility of being useful, because in its lightest processing version (with 125 megabytes), it was what was possible to use in tests with a ready-made example.

2.4 Initial experiments with Open IE based NER

Opening a new line of work - exploration of OpenIE for NER - we started the work with the REBEL OpenIE system [13], using it as an initial extractor of syntactic components.

We are still developing this new process, having recently completed the pipeline. The first very preliminary tests were promising. We will bring more details of the process in the results and conclusions.

Chapter 3

Results

This chapter highlights the main results obtained last year. They integrate presentations, publications and software developments.

3.1 Publications and Presentations

During the period one submission to international conference was accepted, and presented.

3.1.1 IberSpeech 2022

I presented in person the “Assessing Transfer Learning and automatically annotated data in the development of Named Entity Recognizers for new domains” at Iberspeech Conference in Granada, November 2022[9].

Abstract: With recent advances Deep Learning, pretrained models and Transfer Learning, the lack of labeled data has become the biggest bottleneck preventing use of Named Entity Recognition (NER) in more domains and languages. To relieve the pressure of costs and time in the creation of annotated data for new domains, we proposed recently automatic annotation by an ensemble of NERs to get data to train a Bidirectional Encoder Representations from Transformers (BERT) based NER for Portuguese and made a first evalua-

tion. Results demonstrated the method has potential but were limited to one domain. Having as main objective a more in-depth assessment of the method capabilities, this paper presents: (1) evaluation of the method in other domains; (2) assessment of the generalization capabilities of the trained models, by applying them to new domains without retraining; (3) assessment of additional training with in-domain data, also automatically annotated. Evaluation, performed using the test part of MiniHAREM, Paramopama and LeNER Portuguese datasets, confirmed the potential of the approach and demonstrated the capability of models previously trained for tourism domain to recognize entities in new domains, with better performance for entities of types PERSON, LOCAL and ORGANIZATION.

The paper was presented as a poster (in Figure 3.1). Its presentation gave me my first experience in the scientific community of my PhD topic in person, as the previous ones, due to the COVID pandemic, were virtual. So I can say that this contact awakened me to the need to adjust the work, deepening both the knowledge about NER issues and the need to provide better tools to work on entity identification models. Based on the presentation at the conference and the exchange of experiences with other scientists, I believe that working based on machine learning and focusing on the BERT model would be of great value.

The paper also presents examples of new NERs that can be rapidly created with the obtained annotated data. The annotated data, combined with the possibility to apply both the ensemble of NER systems and the new Gazetteer-based NERs to large corpora, create the necessary conditions to explore the recent neural deep learning state-of-art approaches to NER (ex: BERT) in domains with scarce or nonexistent data for training.

3.1.2 New paper in preparation

We are working on developing a new paper to report the results we are getting using Open IE with a focus on creating automatic NER annotation.

3.1.3 Doctoral Consortium

In September 2023 we prepared and submitted application to the Doctoral Consortium of the 15th International Conference on knowledge management and infor-

Assessing Transfer Learning and automatically annotated data in the development of Named Entity Recognizers for new domains.

Emanuel Matos^{1,2}, Mário Rodrigues^{2,3}, António Teixeira^{1,3}

¹FEETA, DETI, University of Aveiro, Aveiro, Portugal

²FEETA, ESTGA, University of Aveiro, Aveiro, Portugal

³IASI - Intelligent Systems Associate Laboratory, Portugal
(easm.mjr.ajstf@ua.pt)



Introduction

The identification of entities is a key step in many natural language processing (NLP) tasks and consequently named entity recognition (NER) is a task important in several contexts, text genres, and languages.

State-of-the-art solutions are based on deep learning models, requiring annotated data for training.

Problems:

➤ The creation of those datasets is time-consuming, costly, and prone to human error.

➤ The lack of labeled data prevents NER being effectively used in some domains and tasks

Objective: Avoid the traditional manual labeling procedure to develop NER for new domains

Proposal

Process consisting of two parts:

1. Cross-domain application of a BERT-based NER model previously trained with automatically annotated data (top part of the Figure);
2. Fine-tuning of this base model to a new domain resorting to automatically annotated data for that domain.

For this initial exploration of cross-domain BERT-base NERs, the previously explored domain of Tourism was selected and complemented with the Legal domain.



Datasets

- **Wikivoyage** Tourism dataset
 - Resulting from automatic annotation of texts from Wikivoyage.
- **LeNER** Legal texts dataset
 - 60 texts from LeNER dataset of legal documents in Brazilian Portuguese.
- **Paramopama** datasets
 - Extension of the PIBR version of Wikiner corpus.
 - Has a total of 240,755 words tagged as part of an entity, considering 4 types (PERSON, LOCATION, ORGANIZATION and TIME).
- **HAREM** datasets
 - Two HAREM datasets were used, the First HAREM and the MiniHAREM, both having manually annotated entities.

References (previous work)

- E. Matos, M. Rodrigues, P. Miguel, and A. Teixeira. "Towards Automatic Creation of Annotations to Foster Development of Named Entity Recognizers" in: ISJATE 2021. Available at: <https://ojs.aaai.org/ocs/viewdoc/download?doi=10.1.1.111.1111>

- E. Matos, M. Rodrigues, and A. Teixeira. "Named Entity Extractors for New Domains by Transfer Learning with Automatically Annotated Data" International Conference on Computational Processing of the Portuguese Language (PROPOR), Springer, Cham, 2022.

Related Work

NER, as many tasks in NLP, has been approached in two very different ways: systems based in rules and lists (the so-called Gazetteers) and data-driven systems base in machine learning. Machine Learning approaches to NER can be more flexible but they depend on the existence of adequate datasets for the target domain.

Several machine learning methods have been applied to NER:

For example, Support Vector Machines (SVM), Conditional Random Field (CRF) or Neural networks (NN).

State-of-the-art NER systems for Portuguese are BERT-based.

Recently, the authors developed NER systems for PT based on BERT using automatically annotated data.

Adopting Transfer Learning, fine-tuning pretrained BERT models with an automatically annotated dataset for the Tourism domain, based in Wikivoyage texts. Best F1 obtained was 64.9 %

Results

The next Table presents the results for the selected metrics as function of the NER model and test-set. For easier understanding, part of the information in the Table is presented graphically in Figures.

Domain	Parameters			LeNER test set			MiniHAREM		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
No Adaptation	60.33	44.33	50.43	59.04	20.17	30.07	62.43	33.98	48.26
20 % of LeNER	68.75	58.37	58.47	58.32	33.38	37.63	63.25	30.58	45.89
40 % of LeNER	61.67	38.51	27.98	58.67	3.89	7.12	66.12	33.92	50.14
60 % of LeNER	61.23	30.72	45.97	68.67	4.20	7.90	64.93	38.44	54.72
80 % of LeNER	61.00	31.36	48.42	62.86	7.74	13.78	63.17	49.24	64.43
100 % of LeNER	60.68	22.06	35.71	76.79	1.59	3.12	66.87	24.49	39.10

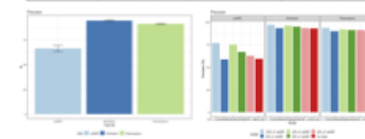


Table and plots show that:

- The results obtained for LeNER legal dataset are worst than the ones obtained for the two other test sets
 - being particularly worst in terms of Recall
- The results for Paramopama are similar to the MiniHAREM results, confirming in this new dataset the results reported in figures
- Despite showing some reduced contribute to improve Precision, the additional training (fine-tuning) of the model with LeNER train dataset resulted in degradation of Recall, for all 3 test sets.
 - The best Recall is obtained without additional training for the 3 datasets, being the best overall score obtained with MiniHAREM.

Conclusions

Addressing the problem of development of NERs for new domains without annotated datasets, the potential of cross-domain application of BERT-based NER models with and without tuning with in-domain additional training was investigated.

As expected, models had a much higher difficulty in recognizing entities of classes not present in the training dataset(s).

Nevertheless, the Recall of approx. 10% for both LEGISLATION and JURISPRUDENCE in LeNER without any in-domain training is an interesting result, with potential to be explored in bootstrap methods.

Additional training with a small amount of in-domain data (automatically annotated) was not very useful.

Despite some improvement in Precision (e.g., for PERSON in LeNER), it contributed to degradation of Recall.

ACKNOWLEDGEMENTS

This research was supported by FEDER Research Unit, funded by National Funds through the FCT - Foundation for Science and Technology, in the context of the project UIDB/01012/2020.



Figure 3.1: The poster presented at IberSpeech 2022 [9].

mation systems, which will take place in Rome between 15th and 17th November. We are awaiting the result.

3.2 New pipeline using Open IE

The Open IE study started with REBEL[13] technology for the initial extraction of entities, then we did a mapping with Wikimapper¹ and after this part we evaluated the 10 most found, these being the chosen entities, as the first possibility. Then, we used the BERT Algorithm to create a model based on training and subsequent evaluation through tests extracted from that same domain. The process developed is shown the Figure 3.2 and examples of its main steps results are presented in Figures 3.3 to 3.6.

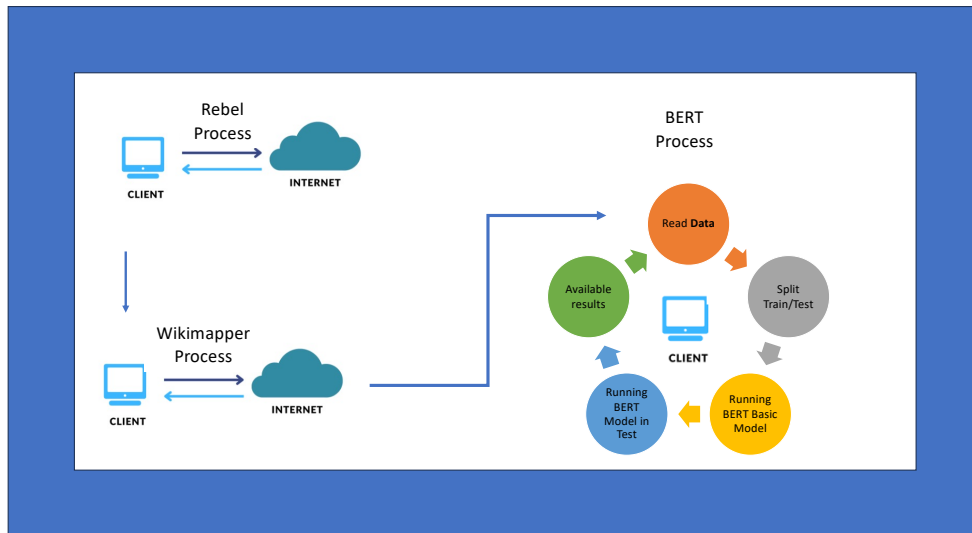


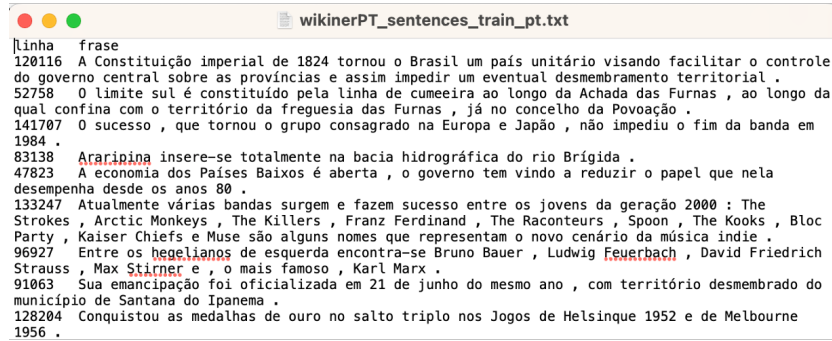
Figure 3.2: Caption From [10]

We are still evaluating the entire pipeline, but it looks promising Also, this form

¹<https://pypi.org/project/wikimapper/>

of automatic creation of entity annotation.

Despite the option made for not developing at this stage the “Tools for non-programmers”², several scripts were developed and existing ones improved to rearrange the process.

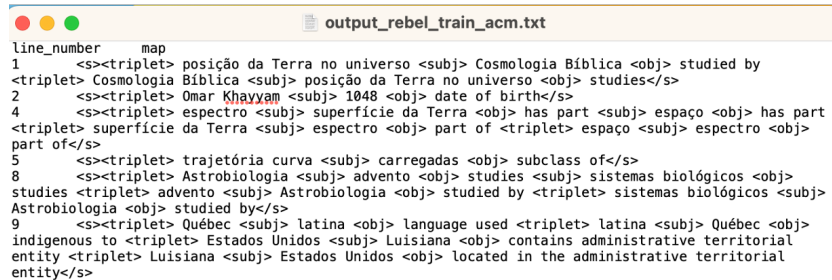


```

linha  frase
120116 A Constituição imperial de 1824 tornou o Brasil um país unitário visando facilitar o controle
do governo central sobre as províncias e assim impedir um eventual desmembramento territorial .
52758 O limite sul é constituído pela linha de cumeeira ao longo da Achada das Furnas , ao longo da
qual confina com o território da freguesia das Furnas , já no concelho da Povoação .
141707 O sucesso , que tornou o grupo consagrado na Europa e Japão , não impediu o fim da banda em
1984 .
83138 Araripina insere-se totalmente na bacia hidrográfica do rio Brígida .
47823 A economia dos Países Baixos é aberta , o governo tem vindo a reduzir o papel que nela
desempenha desde os anos 80 .
133247 Atualmente várias bandas surgem e fazem sucesso entre os jovens da geração 2000 : The
Strokes , Arctic Monkeys , The Killers , Franz Ferdinand , The Raconteurs , Spoon , The Kooks , Bloc
Party , Kaiser Chiefs e Muse são alguns nomes que representam o novo cenário da música indie .
96927 Entre os hegelianos de esquerda encontra-se Bruno Bauer , Ludwig Feuerbach , David Friedrich
Strauss , Max Stirner e , o mais famoso , Karl Marx .
91063 Sua emancipação foi oficializada em 21 de junho do mesmo ano , com território desmembrado do
município de Santana do Ipanema .
128204 Conquistou as medalhas de ouro no salto triplo nos Jogos de Helsinque 1952 e de Melbourne
1956 .

```

Figure 3.3: Example of Text from WikiNER, showing several lines, each with a line number and sentence.



```

line_number  map
1  <s><triplet> posição da Terra no universo <subj> Cosmologia Bíblica <obj> studied by
<triplet> Cosmologia Bíblica <subj> posição da Terra no universo <obj> studies</s>
2  <s><triplet> Omar Khayyam <subj> 1048 <obj> date of birth</s>
4  <s><triplet> espectro <subj> superfície da Terra <obj> has part <subj> espaço <obj> has part
<triplet> superfície da Terra <subj> espectro <obj> part of <triplet> espaço <subj> espectro <obj>
part of</s>
5  <s><triplet> trajetória curva <subj> carregadas <obj> subclass of</s>
8  <s><triplet> Astrobiologia <subj> advento <obj> studies <subj> sistemas biológicos <obj>
studies <triplet> advento <subj> Astrobiologia <obj> studied by <triplet> sistemas biológicos <subj>
Astrobiologia <obj> studied by</s>
9  <s><triplet> Québec <subj> latina <obj> language used <triplet> latina <subj> Québec <obj>
indigenous to <triplet> Estados Unidos <subj> Luisiana <obj> contains administrative territorial
entity <triplet> Luisiana <subj> Estados Unidos <obj> located in the administrative territorial
entity</s>

```

Figure 3.4: Example of output from REBEL processing, showing the tags added to mark the triplets and their parts.

²“Tools for Dummies” in the initial proposal.

line	mapper	base	result
1	Q1845	Bíblia	texto religioso
2	Q19359	1048	ano
4	Q16608018	Espectro	página de desambiguação da Wikimedia
4	Q87919178	Espaço	página de desambiguação da Wikimedia
5	Q413088	Carregada	página de desambiguação da Wikimedia
6	Q2	Terra	região geográfica
8	Q83012	Advento	Q3517779
8	Q411	Astrobiologia	especialidade
8	Q7628051	Studie	negócio
9	Q255932	Latina	página de desambiguação da Wikimedia
9	Q176	Québec	provincia do Canadá
9	Q1588	Luisiana	estado dos Estados Unidos
9	Q30	Estados Unido	país
15	Q360467	Madalena	página de desambiguação da Wikimedia
15	Q739	Colômbia	país
15	Q131792	Orinoco rio	
15	Q717	Venezuela	país
17	Q155	Brasil	país
17	Q664609	Caribe	arquipélago
18	Q96	México	país
20	Q717	Venezuela	país
20	Q155	Brasil	país
20	Q739	Colômbia	país
21	Q790	Haiti	país
22	Q15	África	continente
22	Q155	Brasil	país
23	Q33742	Idioma	Q20829075
23	Q241649	Espanhol	página de desambiguação da Wikimedia

Figure 3.5: Example of output from Wikimapper process.

tag_rebel_train				
Sentence #	Word	Tag	POS	linha
Sentence :1	Bíblia	O	X	1
Sentence :1	Cosmologia_Bíblica	O	X	1
Sentence :2	1048	ENTITY	X	2
Sentence :2	Omar_Khayyam	ENTITY	X	2
Sentence :4	Espectro	O	X	4
Sentence :4	Espaço	O	X	4
Sentence :5	Carregadas	O	X	5
Sentence :8	Advento	O	X	8
Sentence :8	Astrobiologia	O	X	8
Sentence :8	Studie	O	X	8
Sentence :9	Latina	O	X	9
Sentence :9	Québec	O	X	9
Sentence :9	Luisiana	ENTITY	X	9
Sentence :9	Estados_Unidos	ENTITY	X	9□

Figure 3.6: Example of output from BERT process. Column Tag shows the output of the BERT-based NER. The system marks as entity "1048", "Omar Khayyam", "Luisiana", "Estados Uniddos" but fails in many others (ex. "Bíblia").

As is still very manual the process of joining scripts, we believe to create more reliability in process will be better than "easy to use" in this moment. Before

making it more “friendly” to the end user, we still have much to develop and validate, so we consider the overall system still in embryonic stage.

3.3 Comments

The publications planned for the year 22/23 were achieved.

Advances were made in 2023 in datasets available for experiments, processing pipeline and new ways to create automatic of NER systems.

Work as deviated from the plan regarding the initial idea of making a initial system with basic modules that would be later improved. Work is now focused on NER and a more bottom-up approach, developing modules with good individual performance. This new focus will probably imply a request for alteration of the Thesis title in the future.

Chapter 4

Future work

Taking in consideration our results, our goal for 2023-2024 was changed, but the main focus remained, that is, to study and create the building blocks to automate the capture and tagging of entities (NER). We make adjustments to move towards our objective, the next steps considered essential for the development of the work are:

- **State-of-the-art update.**
- **OpenIE-based NER.**
- **NER for new domains.**
- **Custom class sets in NER.**

We organize the items above so that we can have more focus on our work and deliver results that contribute to the scientific development of the area. This reorganization does not harm the initial and basic proposal of our work. This reorganization helps us achieve our purpose more precisely. This reorganization is reflected in the updated work plan.

4.1 Updated Workplan

Based on what happened in 2022/2023, working part-time and with many daily interferences, we believe that the adjustments to the plan we propose will result in more consistent work and academic contribution, because the experience of the years combined with the support of supervisors created a balance that it makes sense to make adjustments for the next periods. The reorganized is presented in table 4.1.

Task	2023-2024				2024-2025			
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q
State-of-the-art update		X	X	X	X	X	X	
OpenIE-based NER ¹	X	X	X	X				
NER 4 new domains ²		X	X	X	X	X		
custom class sets in NER ³		X	X	X	X	X		
Publications	X		X		X		X	
Thesis	X	X	X	X	X	X	X	X

Table 4.1: Revised Workplan.

Despite the extension of the time required, 2023 showed us that the balance between academic life and the professional was hard but still makes a gratefully time, as they are synergistic, as long as we have resilience in everyday life.

4.1.1 Milestones - Update

According to the changes made to the workplan, Milestones were also revised, as follows:

- M1 - delivery of the “NER without data registration”. Planned to the end of the 1st complete year of Thesis work, was achieved a bit later, in the final months of 2021.
- M2 - New conceptions of NER classifications (including OpenIE-based NER). Planned for 4th quarter of 2022/2023, is work in progress. New

¹RQ: OpenIE can provide a NER useful to create automatic annotations?

²RQ: our approach generalizes to other domains?

³RQ: Can we go beyond PERSON, LOC, ORG classic class sets with automatically annotated data?

pipeline needs further analysis and development. New target is 2nd quarter 2023/2024.

- M3 - NER for custom class sets (and new domains) Resulting directly from the revised workplan and planned for the 1st Quarter of 24/25.
- M4 - Document detailing the work carried out and the scientific results obtained to support the proposed research questions. Planned for September 2025.

Chapter 5

Conclusion

Supported by analysis of the results and work evolution, we propose adjustments to some points of the initial proposal, without compromising its base. On the contrary, we believe that this new, more informed, vision was only possible through the studies that took place during years. Support that the work is progressing in the right direction was provided by the feedback to the paper and poster presentation at IberSpeech in Granada.

In line with our overall goals, the plan for 2024 to 2025 has been adjusted, we will focus on development of core modules and creation of NER for new domains, and then, if possible, we will create applications aggregating the parts. In result, the tools for non-developers will not be part of next year objective.

A major result for this year was the development of an updated list of Research questions (RQ), that will guide all the work for the next 2 years. They are directly related to Tasks in the revised workplan (and included in it as footnote):

RQ1: Can we produce NERs with reasonable performance with automatically annotated data?

RQ1b: Can we create useful annotated data combining existing systems?

RQ2: Open IE can provide a NER useful to create automatic annotations?

RQ3: Can we go beyond classic class sets with automatically annotated data?

RQ4: Approaches generalize to other domains?

Bibliography

- [1] K. Sintoris and K. Vergidis. Extracting business process models using natural language processing (NLP) techniques. In *2017 IEEE 19th Conference on Business Informatics (CBI)*, volume 01, pages 135–139, 2017.
- [2] Flora Amato, Giovanni Cozzolino, Vincenzo Moscato, and Francesco Moscato. Analyse digital forensic evidences through a semantic-based methodology and nlp techniques. *Future Generation Computer Systems*, 98:297–307, 2019.
- [3] Antonio Moreno Sandoval, Julia Díaz, Leonardo Campillos Llanos, and Teófilo Redondo. Biomedical term extraction: Nlp techniques in computational medicine. *IJIMAI*, 5(4):51–59, 2019.
- [4] Hiral Desai, Mohammed Firdos Alam Sheikh, and Satyendra K Sharma. Multi-purposed question answer generator with natural language processing. In *Emerging Trends in Expert Applications and Security*, pages 139–145. Springer, 2019.
- [5] Joel Nothman, Nicky Ringland, Will Radford, Tara Murphy, and James R Curran. Learning multilingual named entity recognition from wikipedia. *Artificial Intelligence*, 194:151–175, 2013.
- [6] António Teixeira, Pedro Miguel, Mário Rodrigues, José Casimiro Pereira, and Marlene Amorim. From web to persons - providing useful information on hotels combining information extraction and natural language generation. In *Proc. IberSpeech*, Lisbon, November 2016.
- [7] Wikivoyage. <https://pt.wikivoyage.org/>.
- [8] C Mendonça Júnior, Hendrik Macedo, Thiago Bispo, Flávio Santos, Nayara Silva, and Luciano Barbosa. Paramopama: a brazilian-portuguese corpus for

- named entity recognition. *Encontro Nac. de Int. Artificial e Computacional*, 2015.
- [9] Emanuel Matos, Mário Rodrigues, and António Teixeira. Assessing Transfer Learning and automatically annotated data in the development of Named Entity Recognizers for new domains . In *Proc. IberSPEECH 2022*, pages 191–195, 2022.
- [10] Emanuel Matos, Mário Rodrigues, Pedro Miguel, and António Teixeira. Towards automatic creation of annotations to foster development of named entity recognizers. In *10th Symposium on Languages, Applications and Technologies (SLATE 2021)*. Schloss Dagstuhl-Leibniz-Zentrum für Informatik, 2021.
- [11] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.
- [12] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: pre-training of deep bidirectional transformers for language understanding. *CoRR*, abs/1810.04805, 2018.
- [13] Pere-Lluís Huguet Cabot and Roberto Navigli. Rebel: Relation extraction by end-to-end language generation. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 2370–2381, 2021.